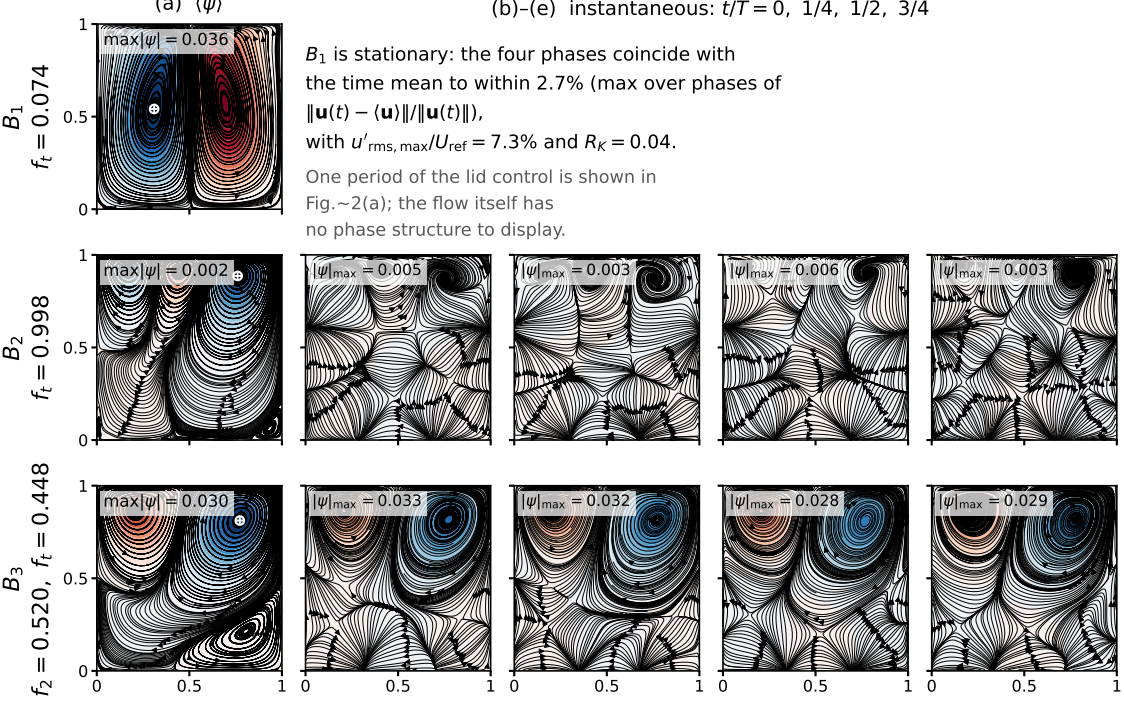

(a) $\langle\psi\rangle$
(b)–(e) instantaneous: $t/T = 0,\ 1/4,\ 1/2,\ 3/4$
B_1 is stationary: the four phases coincide with the time mean to within 2.7% (max over phases of $\|\mathbf{u}(t) - \langle\mathbf{u}\rangle\|/\|\mathbf{u}(t)\|$), with $u'_{\mathrm{rms,max}}/U_{\mathrm{ref}} = 7.3\%$ and $R_K = 0.04$.
One period of the lid control is shown in Fig.~2(a); the flow itself has no phase structure to display.
B_1 $f_t = 0.074$
B_2 $f_t = 0.998$
B_3 $f_2 = 0.520$, $f_t = 0.448$
$\max|\psi| = 0.036$
$\max|\psi| = 0.002$
$|\psi|_{\max} = 0.005$
$|\psi|_{\max} = 0.003$
$|\psi|_{\max} = 0.006$
$|\psi|_{\max} = 0.003$
$\max|\psi| = 0.030$
$|\psi|_{\max} = 0.033$
$|\psi|_{\max} = 0.032$
$|\psi|_{\max} = 0.028$
$|\psi|_{\max} = 0.029$
0
0.5
1
one scale and one level set for all eleven field panels
streamlines: same density and placement rule
−4
0
4
stream function $\psi/(U_{\mathrm{ref}})$ $[\times 10^{-2}]$